Exercise 01: Product Inventory

Design a `Product` class for a simple inventory system.

- A product has a name and a stock quantity.
- The stock quantity must not be changed directly from outside the class.
- The stock can only be changed through valid business operations.

The product should support:

- Add Stock — increase the available stock by a given quantity.
- Sell — reduce the stock when a given quantity of a product is sold.
- A product cannot be sold if sufficient stock is not available.
- Zero or negative quantities should not be accepted.
- The current stock should be available from outside the class.

Design the class so that a `Product` object can always protect itself from entering an invalid state.

What Is Expected from You

In this practice, you are expected to demonstrate your ability to:

- Design a class with appropriate data and behaviors.
- Follow proper naming conventions.
- Provide multiple ways to initialize an object when appropriate.
- Keep the internal state (data) private.
- Apply encapsulation properly—not just getters and setters.
- Ensure the object protects its own state, so callers cannot put it into an invalid state.